Google Cloud

Components of BigQuery table schema

So far, you've been introduced to data models and schemas. Data models are tools for organizing data elements and understanding how they relate to one another. Schemas describe how that data is organized. They also define rules and relationships within data, and are key for ensuring data consistency and integrity. Schemas maintain a uniform structure to ensure that the database is easy for users to navigate. This leads to more efficient queries. BigQuery also relies on schemas to describe how data in tables is organized. In this reading, you'll learn more about table schemas in BigQuery, and their components.

Schema components

When creating a new table in BigQuery, you need to specify the schema. This means addressing the specific schema components so the database can organize the data correctly. BigQuery will usually auto-detect the schema of uploaded data files, but understanding the different schema components can ensure that the schema is being applied correctly.

Column names

Column names represent the columns in a table. To successfully query your table, these column names need to be correctly defined and follow GoogleSQL rules. Column names must be less than 300 characters, and should not include any duplicates. It's important to note that there are a few column name prefixes that will cause your code to throw an error. These prefixes include:

- **_TABLE_**
- **_FILE_**
- **_PARTITION**
- **_ROW_TIMESTAMP**
- **__ROOT__**
- **_COLIDENTIFIER**

Column data type

Another important component of your table schema is the column data type. Defining the column type allows the database to recognize what type the data is. This also allows you to transform the data, and use it for calculations and analysis more easily. Review the table to learn about the common data types you'll likely assign:

Data type	SQL	Definition
Integer	INT64	Numeric values without fractional components
Floating point	FLOAT64	Numeric values with fractional components
Numeric	NUMERIC	Exact numeric values with fractional components
BigNumeric	BIGNUMERIC	Exact numeric values with fractional components
Boolean	BOOL	TRUE or FALSE
String	STRING	Character text data
Bytes	BYTES	Binary data
Date	DATE	Calendar date
Date/Time	DATETIME	Year, month, day, hour, minute, second, and subsecond
Time	TIME	Time independent from date
Timestamp	TIMESTAMP	Absolute point in time
Struct (Record)	STRUCT	Container of ordered fields each with a type and field name
Geography	GEOGRAPHY	A pointset on Earth's surface
JSON	JSON	Represents JSON

Default values

Along with defining the column names in a table schema, you can also include a default value for columns. This will allow you to assign a predetermined value to a column within your table. These default values must be either a literal or a specific function. They include:

- CURRENT_DATE
- CURRENT_DATETIME
- CURRENT_TIME
- CURRENT_TIMESTAMP
- GENERATE_UUID
- RAND
- SESSION_USER
- ST_GEOGPOINT

Key takeaways

As you become more familiar with BigQuery, and begin to create your own tables to query, understanding schemas will be key to generating functional useful tables. Cloud data professionals need to understand data modeling and schemas to effectively store, manage, and analyze data for their organizations. Throughout this course, you can reference these schema components to guide your own table creation.